Theraband Anchor

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Design Problem

Our task is to design and construct a low-cost and portable device to act as an anchor for Therabands (resistance bands for exercise) in order for users to conduct a wide range of physical therapy exercises at different heights and weights.



Figure 1: Current Theraband Usage

Motivation

- Therabands are small, low cost, portable elastic bands used for physical rehabilitation and exercise
- To safely conduct exercises with Therabands, it's necessary to anchor it down
- Current anchors either require hardware to install, are expensive, or not completely safe
- Many people can't afford to purchase or install permanent anchors
- Users who are in a wheelchair may not be able to adjust the anchor without the assistance of another person
- Injuries can result from the use of an unreliable anchor

Design Criteria

Objectives	Target Value
Low Cost	≤ $20
Portability	Moved > 100 times without hardware
Adjustability	Adjustable between 0 and 7 feet
Ease of Use	Average score ≥ 4 on user defined scale
Sanitation	Cleaned > 100 times
Constraints	
Safety	Slide < 2 inches without breaking when 35lbs of force is applied > 100 times

Results

Our device is a ratchet strap that goes around a door vertically with evenly spaced loops on one side. Users will ratchet down the strap so it is tight, put the Theraband through a carabiner, and attach the carabiner to a loop in order to perform an exercise.

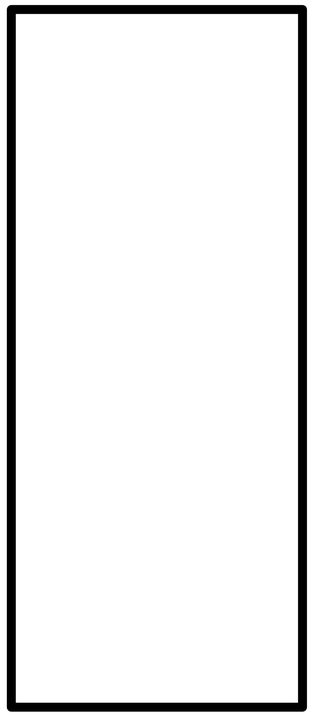


Figure 1: Front view

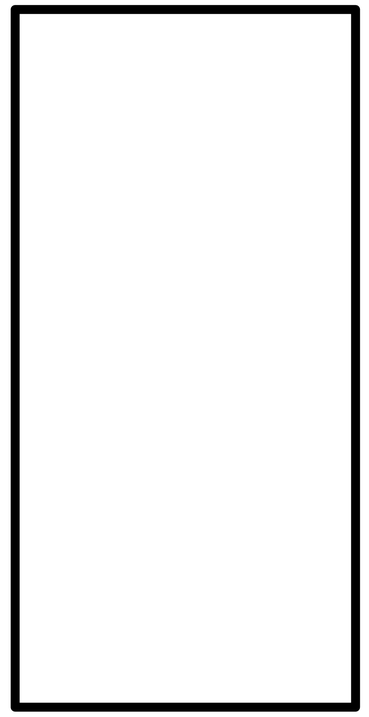


Figure 2: Back view

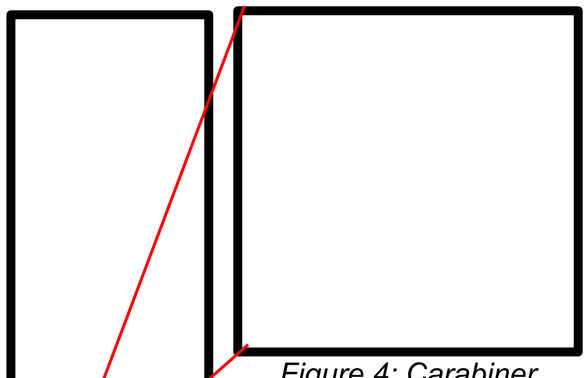


Figure 4: Carabiner

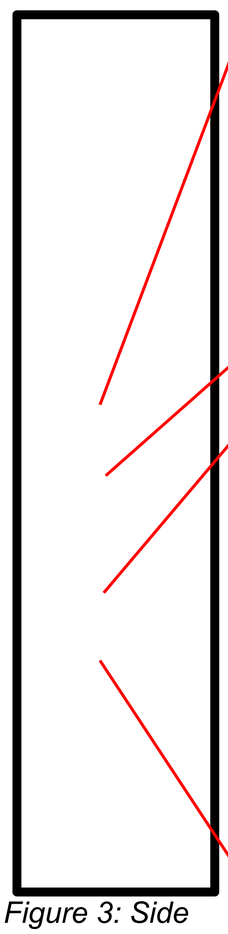


Figure 3: Side view

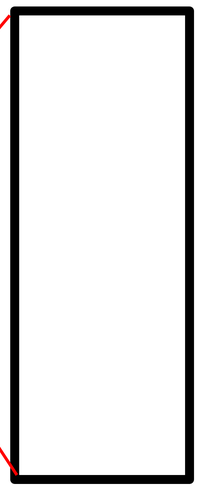


Figure 5: Loop

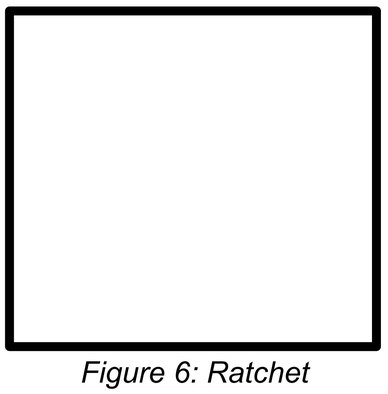


Figure 6: Ratchet strap

Testing

Objectives	Test	Result
Low Cost	Calculate the cost of the prototype	PASS: Cost = $12.64
Portability	Install prototype 100 times without hardware with no visible damage	PASS
Adjustability	Adjust prototype between all loops 10 times for 10 workouts without failure	PASS
Ease of Use	Have user complete a survey based on a user defined scale after completing a workout	PASS: Average = 4.35
Sanitation	Clean prototype 100 times without visible damage	PASS
Safety	Slide < 2 inches without breaking when 35lbs of force is applied > 100 times	PASS

Conclusion

We successfully created a device that allows the anchoring of a Theraband that is both adjustable and portable. Our device is low cost and easy to use.

We hope to implement our design not only for physical therapy needs at Duke Health, but also to have the device available for personal use.

Future Work:
Research slightly less bulky and more intuitive ratchets to go with the strap designs.

References

Amanda Beaty, DPT, PT, Duke Health, Client
Kevin Caves, Duke University, Professor
Jacklyn Herzberg, Duke University, Teaching Assistant
Dr. Roarke Horstmeyer, Duke University, Professor
Bruce Klitzman, Ph.D., Duke University, Technical Mentor